JAYESH BACCHAV

Junior AI Data Engineer

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PROFESSIONAL SUMMARY

Early-career AI/Data Engineer with hands-on experience in **Machine Learning, NLP, and Data Analytics**, with strong foundations in **Python and SQL**. Proficient in building **LLM-based Applications, Data Pipelines, and Predictive Models** using **LangChain and Power BI**. Passionate about leveraging **AI and Data-Driven Solutions** to solve real-world problems and continuously learning **Emerging Technologies**.

SKILLS

• **Programming**: Python, SQL
• **Data Engineering**: Data Pipelines, ETL Workflows, Data Modeling, Data Architecture, Data Transformation, Data Quality
• **Machine Learning & AI**: ML Algorithms, Feature Engineering, Feature Selection, Handling Imbalanced Data, Model Evaluation, Deep Learning (LSTM, NN), NLP
• **Generative AI & LLMs**: LangChain, Prompt Engineering, Agent Systems, RAG (basic), Llama3
• **Analytics & Libraries**: EDA, KPI, Statistical Analysis, Power BI (DAX), Pandas, NumPy, Scikit-learn, Matplotlib
• **Tools & Databases**: Git, GitHub, Power BI, MySQL, PostgreSQL

PROJECTS

Payment Intelligence Chat Agent (AI-Based Fraud Detection System)

- Built an **LLM-powered chat agent** using *LangChain and Llama3* to interpret user queries and route them to SQL-based analytics or ML-based prediction workflows, improving accessibility of fraud insights for non-technical users.
- Designed a **modular system architecture** with decision (LLM), data (SQL engine), and prediction (ML model) layers, enabling scalable and efficient system integration.
- Developed a **fraud risk prediction model** using transactional data, applying feature engineering, data preprocessing, and handling imbalanced datasets to improve detection of high-risk transactions.
- Implemented a SQL-driven analytics engine to perform data retrieval, aggregation, and filtering on historical transaction data, enabling faster identification of suspicious merchant patterns.
- Enabled a **unified system** combining **descriptive and predictive analytics**, allowing users to identify **high-risk merchants** and gain **actionable fraud insights**.

Feedback Analysis System using LSTM & GPT (Final Year Project)

- Developed a **feedback analysis system** leveraging *LSTM-based sentiment classification* to process and categorize student feedback and Integrated **GPT-based recommendation generation** to convert sentiment insights into actionable suggestions for improving teaching outcomes.
- Designed a **modular architecture** with feedback collection, analysis, and suggestion modules for streamlined processing and scalability.
- Enabled teachers to **identify sentiment trends and take targeted actions** to improving feedback utilization and teaching effectiveness.

HR Analytics & Workforce Planning (SQL & Power BI)

- Built **SQL-based data pipelines** using **CTEs** and **Window Functions** to analyze employee attrition, performance and tenure patterns.
- Integrated **multi-source workforce data** to uncover key trends and generate actionable insights for HR decision-making.
- Developed **interactive Power BI dashboards** with DAX-driven KPIs to monitor employee lifecycle and organizational metrics.
- Enabled identification of **high-risk attrition segments**, supporting data-driven retention strategies and workforce planning.

EDUCATION

Bachelor of Engineering – Artificial Intelligence & Data Science (CGPA: 7.48)
MET BKC Institute of Engineering, Nashik (2021 – 2025)

CERTIFICATIONS

IBM Data Science Professional Certificate | Zensar Python Developer | AI Operations Training